Buying New, Buying Second-Hand, or Renting?

A Life-Cycle and AHP-Based Decision Analysis of Formalwear for Students

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Abstract

Students choosing whether to purchase or rent formalwear must balance carbon emissions, expenditure, and convenience. This study compares purchasing a new dress, purchasing a second-hand dress, and renting the same dress model for a National University of Singapore student attending six formal events over one year. Life cycle assessment (LCA) estimates allocated annual greenhouse gas emissions, while life cycle costing (LCC) quantifies direct consumer expenditure. These outcomes are combined with a time-based convenience indicator through analytic hierarchy process (AHP)-weighted composite scoring. Under baseline assumptions, allocated annual emissions are 4.92 kg CO_2e for each purchase option and 11.38 kg CO_2e for rental. Annual expenditure is SGD 60.99, 72.00, and 108.00 for second-hand purchasing, new purchasing, and rental, respectively. Second-hand purchasing ranks first across the evaluated weighting specifications under baseline operating assumptions, reflecting its lower expenditure and identical modelled emissions and active time requirements relative to new purchasing. Transport sensitivity analysis further examines how collection-and-return travel burdens affect the ranking of the three alternatives. The findings support second-hand purchasing in the baseline scenario and identify substantial reductions in attributed collection-and-return travel burdens as a condition for rental competitiveness within the modelled scenarios.

Keywords: formalwear; life-cycle assessment; consumer expenditure; analytic hierarchy process; group judgments; transport sensitivity

1 Introduction

Students attending formal events select among garment ownership, second-hand acquisition, and temporary access. Each arrangement combines an acquisition payment with cleaning, transport, and personal effort. The environmental performance of repeated garment use depends on lifetime utilization and the operations required to deliver the clothing service. A comparison on a common annual service basis can clarify how these components influence the preferred acquisition strategy.

This study examines the Love, Bonito Eyka Multi-way Satin Column Dress in size S across three documented Singapore acquisition channels. The research question is: *For one student attending six formal events in one year, how do buying new, buying second-hand, and renting compare in allocated greenhouse gas emissions, direct expenditure, and an AHP-weighted decision score?* The alternatives provide the same model and permit repeated appearance. The comparison assumes adequate fit, dress-code suitability, hygiene, and availability.

The analysis evaluates three fixed acquisition strategies through a reproducible sequence of evidence classification, annual burden calculation, preference aggregation, performance scoring, and transport sensitivity analysis. The baseline establishes the relationship among environmental, financial, and time outcomes. The survey supports an exploratory account of criterion preferences, while the scenario analysis identifies service conditions associated with changes in ranking.

2 Background and Related Work

2.1 Clothing life cycles and access arrangements

Clothing assessments require a common service definition and explicit allocation of garment production across use. Sandin et al. (2019) document inventories and environmental impacts for six

garment types, including a polyester dress. Their garment-level estimates provide screening proxies for the present analysis. Monticelli and Costamagna (2023) compare formalwear rental and online purchasing in an Italian setting, highlighting the influence of use frequency and service operations. Their residential-care estimate supplies a cleaning proxy for the present study.

The baseline assigns a common lifetime of 26 wears to all channels. This assumption provides a controlled comparison of acquisition, care, and transport arrangements at a common utilization level. The resulting carbon accounts describe attributed annual service burdens. Their interpretation depends on the lifetime and allocation convention, garment inventories, and delivery practices specified in the model.

2.2 Preference aggregation and consistency

AHP translates reciprocal pairwise judgments into criterion priorities and assesses within-matrix consistency (Saaty, 1994). Group applications distinguish the aggregation of individual judgments (AIJ) from the aggregation of individual priorities (AIP). AIJ combines judgments into a collective comparison matrix, whereas AIP combines the priority vectors of respondents treated as individual decision makers (Forman & Peniwati, 1998).

Geometric aggregation can produce an acceptably consistent group matrix when individual matrices exhibit substantial inconsistency (Aull-Hyde et al., 2006). Individual and group consistency consequently describe distinct levels of the judgment process. The present study reports both levels and examines alternative mappings and aggregation procedures. AHP supplies criterion weights; externally specified linear value functions translate the three alternatives' annual burdens into comparable performance scores.

3 Study Design and Methods

3.1 Functional unit, alternatives, and system boundary

The functional unit is suitable formalwear for one student attending six formal events over one year. New and second-hand purchasing each comprise one acquisition followed by six wears. Rental comprises six separate seven-day bookings, each covering one event. Events occur in separate booking periods, and the baseline permits the same style throughout the year. The second-hand listing identifies the selected model and size, with its colour retained as a comparability uncertainty.

The environmental inventory comprises allocated manufacturing, pre-consumer distribution and retail, gross end-of-life disposal, annual cleaning, and acquisition-related delivery or passenger travel. Event travel, shoes, accessories, accidental repairs, and new packaging remain outside this inventory; the packaging assumption is reuse. The distribution factor supplies the retail component. Annual disposal allocation represents a share of eventual end-of-life treatment over the service life.

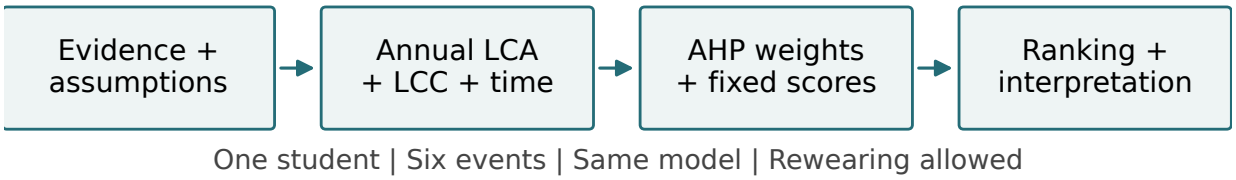


Figure 1: Analytical workflow linking evidence and assumptions to annual burdens, criterion weights, performance scores, and interpretation. The transport scenario varies the acquisition-travel component within this framework.

3.2 Data sources and model inputs

The data register distinguishes advertised market quotes, literature proxies, original survey responses, and researcher-defined scenarios (Table 1). Market values are frozen advertised quotes recorded on 3 November 2026. The official product page specifies a polyester main fabric and polyester–spandex lining (Love, Bonito, n.d.). The second-hand and rental listings identify the same model and size (forest_green66, n.d.; igotgr8deals, n.d.). Garment mass, item condition, and transaction availability remain product-level evidence uncertainties.

Table 1: Principal model inputs and evidence classification.

Input	Baseline value	Evidence and provenance
Acquisition payments	New SGD 65; second-hand SGD 52; rental SGD 10/7 days	Frozen advertised quotes: Love, Bonito and two Carousell listings
Delivery and deposit	New SGD 0; second-hand SGD 1.99; refundable rental deposit SGD 99	Advertised terms; full refund assumed
Manufacturing / distribution / disposal	14.8 / 0.84 / 0.99 kg CO_2e per garment	Rounded graphical extractions from Sandin et al. (2019), Fig. 4.24
Reference garment	0.478 kg polyester dress	Literature reference mass
Cleaning factor	0.109 kg CO_2e per clean	Residential washing, drying, and ironing proxy; Monticelli and Costamagna (2023), Table 4
Rail / van factors	0.0287 kg CO_2e/passenger-km; 0.31844 kg CO_2e/vehicle-km	UK 2026 factors, including direct and upstream energy emissions
Annual / lifetime use	6 / 26 wears, all channels	Common service and utilization scenario
Cleaning counts	Purchases 7; rental 6	Initial wash plus six post-wear washes for ownership; six post-return washes for rental
Acquisition travel	Purchases: 1 allocated vehicle-km; rental: 240 passenger-km	Assumed delivery batch and four dedicated rail legs per booking
Care and fares	SGD 1/home clean; SGD 2/travel leg; rental-care surcharge SGD 0	Research assumptions
Active time	Purchases 120; rental 780 minutes/year	Ordering, care, rental handling, and travel assumptions
Annual score anchors	24 kg CO_2e; SGD 360; 1080 minutes	Reference burdens for linear scoring
Criterion preferences	18 original records; 17 complete matrices	Five-position survey responses and explicit ratio conversions

Note. The reference mass describes the source garment. Emission factors already use a per-garment basis. Source locators appear in Appendix A.

The supplier's care instructions motivate one initial wash for both purchase options. Rental care occurs after return, consistent with the listing's return instructions. The shared care factor represents residential washing, drying, and ironing in an Italian case. Transport factors represent UK rail and van activity (Department for Energy Security and Net Zero, 2026). Their application to Singapore constitutes a geographical proxy assumption. The second-hand seller's reported prior wear enters the remaining-capacity check, $1 + 6 \leq 26$.

3.3 Annual carbon accounting

Let $m \in \{N, S, R\}$ denote new, second-hand, and rental, respectively. With $n = 6$ annual wears, $L = 26$ lifetime wears, and per-garment manufacturing, distribution, and gross disposal factors

M, D, F, annual attributed emissions are

$$E_m = \frac{n}{L}(M + D + F) + k_m e_c + d_m e_v + p_m e_r, \tag{1}$$

where k_m is cleaning count, d_m allocated delivery vehicle-kilometres, p_m passenger-kilometres, and e_c, e_v, e_r the corresponding factors. The baseline uses $(k_N, k_S, k_R) = (7, 7, 6)$, $(d_N, d_S, d_R) = (1, 1, 0)$, and $(p_N, p_S, p_R) = (0, 0, 240)$.

The common garment-stage contribution is $(6/26)(14.8 + 0.84 + 0.99) = 3.8377$ kg CO_2e/year. Production, distribution, and gross disposal are allocated identically across channels. Avoided-production and energy-recovery credits are set to zero. The reference mass serves as descriptive inventory information. The source factors retain their published impact-method conventions, yielding a screening carbon assessment with explicit geographical and methodological transfer assumptions.

3.4 Annual consumer expenditure and active time

The consumer cost model measures direct cash expenditure over the study year. Let P_m denote purchase payment, H_m delivery payment, r_m rental fee per booking, c_h home-care cost, c_x rental-care surcharge, b_m paid passenger legs, and f fare per leg. Then

$$C_m = P_m + H_m + nr_m + \mathbf{1}_{m\in\{N,S\}} k_m c_h + \mathbf{1}_{m=R} n c_x + b_m f. \tag{2}$$

Inapplicable components take a value of zero. Purchase payments enter in full, and resale receipts and year-end residual value are set to zero. This convention measures annual cash affordability, while the carbon model attributes garment burdens to annual service use. The SGD 99 rental deposit is reported separately as a temporary liquidity requirement under the full-refund assumption. Sequential bookings require one deposit at a time. The baseline assumes timely returns in acceptable condition and zero additional platform and ordinary-care charges.

Convenience is operationalized as active consumer time:

$$T_N = T_S = 15 + 7(15) = 120 \text{ minutes/year}, \tag{3}$$

$$T_R = 6[10 + 4(30)] = 780 \text{ minutes/year}. \tag{4}$$

The components comprise ordering, active home care, booking and handover, and dedicated travel. These values are researcher-defined scenarios. The operationalized construct is the active-time component of convenience; broader search, fitting, and psychological effort remain areas for further measurement.

3.5 Survey description and AHP specification

The original questionnaire contains 18 records. Question-level descriptions use the available answers to each item. Seventeen records contain all three criterion comparisons, Q8–Q10; record R017 contributes its two recorded comparisons to the question-level summaries. All survey statistics and preference calculations derive from these original responses.

The criteria are emissions (E), expenditure (C), and convenience (V). Q8 supplies E/C, Q9 supplies E/V, and Q10 supplies C/V. Q8 provides five verbal categories centred on equal importance. Q9 and Q10 provide labelled endpoints and numbered intermediate positions. For these two items, midpoint equality and symmetric intensity form explicit analytical interpretations. The baseline mapping assigns distance from code 3, $|x - 3| = 0, 1, 2$, to ratio intensities $1, 3, 5$. Where code 1 favours the numerator, codes 1–5 map to $(5, 3, 1, 1/3, 1/5)$. Q9 reverses this sequence because its code 1 favours convenience. A milder mapping uses intensities $1, 2, 3$ to examine uncertainty in the conversion of ordered responses to ratios.

For respondent r, the positive reciprocal comparison matrix is

$$A^{(r)} = \begin{pmatrix} 1 & a_{EC}^{(r)} & a_{EV}^{(r)} \\ 1/a_{EC}^{(r)} & 1 & a_{CV}^{(r)} \\ 1/a_{EV}^{(r)} & 1/a_{CV}^{(r)} & 1 \end{pmatrix}. \tag{5}$$

For a 3×3 positive reciprocal matrix, normalized row geometric means equal the principal eigenvector:

$$g_i = \left(\prod_{j=1}^{3} a_{ij} \right)^{1/3}, \qquad w_i = \frac{g_i}{\sum_{h=1}^{3} g_h}. \tag{6}$$

The calculation checks $Aw = \lambda_{\max} w$. Consistency is summarized by

$$CI = \frac{\lambda_{\max} - 3}{2}, \qquad CR = \frac{CI}{RI}. \tag{7}$$

The baseline uses $RI = 0.52$, from the three-element table on printed page 104 of Saaty (1994), and the criterion $CR \leq 0.10$. A parallel calculation uses the course reference value $RI = 0.58$. RI affects consistency diagnostics and CR-based subset membership; priorities calculated from a given matrix depend on its judgments.

For q respondents, AIJ forms the collective matrix as

$$\bar{a}_{ij} = \exp\left(\frac{1}{q} \sum_{r=1}^{q} \ln a_{ij}^{(r)} \right). \tag{8}$$

The baseline AIJ retains all 17 complete responses with equal respondent weights. The comparison includes the milder mapping, an AIP arithmetic average $\bar{w}_i = q^{-1} \sum_r w_i^{(r)}$, and an equal-weight reference $w_i = 1/3$. CR-qualified subsets provide descriptive selection diagnostics. Individual consistency, collective consistency, mapping sensitivity, and aggregation sensitivity are reported separately. These specifications characterize the available preference evidence as exploratory.

3.6 Performance scoring and transport scenarios

For burden $B_{mj} \in \{E_m, C_m, T_m\}$ and annual reference U_j, the linear performance score and weighted composite are

$$s_{mj} = 1 - \frac{B_{mj}}{U_j}, \qquad S_m = w_E s_{mE} + w_C s_{mC} + w_V s_{mV}. \tag{9}$$

The reference burdens are $U_E = 24$ kg CO_2e/year, $U_C = 360$ SGD/year, and $U_V = 1080$ minutes/year. They specify the scales of the value functions. The scoring function retains its linear form across the burden domain, including values beyond the reference anchors. Higher composite scores indicate preferred outcomes. The effective contribution per unit of burden is governed jointly by w_j and U_j.

The transport analysis preserves the six-event functional unit and the three acquisition strategies. Let $\alpha \in [0, 1]$ denote the share of baseline rental collection-and-return travel attributed to the rental activity. The baseline is $\alpha = 1$; the lower boundary $\alpha = 0$ represents fully shared journeys with zero incremental travel burden. Passenger-travel emissions, incremental fares, and travel time scale proportionally with α, while six rental payments, six cleans, and 60 minutes of booking and handover remain fixed:

$$E_R(\alpha) = 4.4916923 + 6.888\alpha, \tag{10}$$

$$C_R(\alpha) = 60 + 48\alpha, \tag{11}$$

$$T_R(\alpha) = 60 + 720\alpha. \tag{12}$$

This linked specification defines a controlled scenario family. Its proportional relationship among emissions, fares, and time is a modelling assumption. Substituting these equations into Eq. 9 gives $S_R(\alpha) = a - b\alpha$. Rental–ownership crossover shares follow from $\alpha^* = (a - S_m)/b$. The analysis repeats this calculation under each weight specification to examine preference sensitivity of the travel threshold.

4 Results

4.1 Original questionnaire description

Table 2 summarizes the original sample, reported activities, rewearing responses, and pairwise choices. All 18 records identify students studying in Singapore. Thirteen of the 17 available Q8 answers favour expenditure reduction over emissions reduction. Q9 contains eight responses on the convenience side, three on the emissions side, and seven at the midpoint. Q10 contains six on the cost side, three on the convenience side, and nine at the midpoint. These counts describe the direction of recorded preferences on each question's own response scale.

Table 2: Descriptive profile and criterion comparisons in the original questionnaire.

Item	Response	n	%
A. Respondent profile ($N = 18$)			
Q1: Student status	Studying in Singapore	18	100.0
Q2: Events in past 12 months	0	8	44.4
	1–2	5	27.8
	3–4	4	22.2
	5–6	0	0.0
	7 or more	1	5.6
Q4: Same-style reuse	Probably willing	6	33.3
	Uncertain	4	22.2
	Probably unwilling	6	33.3
	Definitely unwilling	2	11.1
B. Pairwise response directions			
Q8: Emissions vs. cost	Emissions	1	5.9
	Equal importance	3	17.6
	Cost	13	76.5
Q9: Emissions vs. convenience	Convenience	8	44.4
	Midpoint	7	38.9
	Emissions	3	16.7
Q10: Cost vs. convenience	Cost	6	33.3
	Midpoint	9	50.0
	Convenience	3	16.7

Note. Question-specific denominators use available responses. Q8 has 17 recorded answers; Q9 and Q10 each have 18. Midpoint denotes the central numbered position in Q9 and Q10. Complete-matrix AHP uses 17 records.

4.2 Annual environmental, financial, and time outcomes

Both purchase strategies yield 4.92 kg CO_2e/year. Their stage contributions comprise 3.4154 from manufacturing, 0.1938 from distribution and retail, 0.2285 from gross disposal, 0.7630 from cleaning, and 0.3184 from delivery. Rental yields 11.38 kg CO_2e/year, including the same attributed garment-stage burden, 0.6540 from cleaning, and 6.8880 from rail travel (Figure 2). The rental–purchase difference is 6.4606 kg CO_2e/year: acquisition transport contributes an additional 6.5696, offset by a cleaning reduction of 0.1090.

Annual expenditure is SGD 60.99 for second-hand, SGD 72.00 for new, and SGD 108.00 for

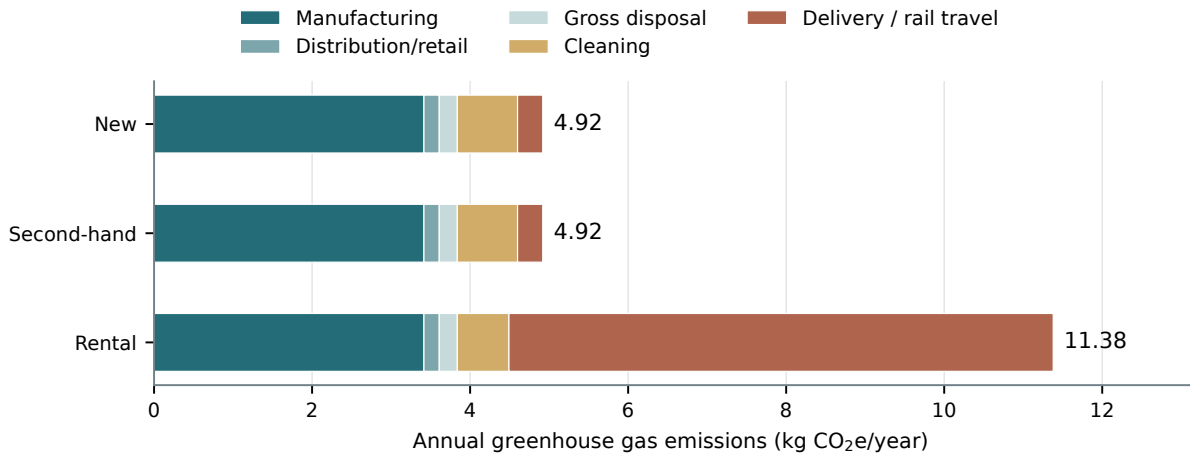


Figure 2: Annual greenhouse gas emissions by life-cycle stage under the common 26-wear allocation. Acquisition transport combines allocated purchase delivery and rental passenger travel.

Table 3: Annual consumer expenditure by component (SGD).

Option	Garment	Delivery	Rent	Care	Travel	Total
New	65.00	0.00	0.00	7.00	0.00	72.00
Second-hand	52.00	1.99	0.00	7.00	0.00	60.99
Rental	0.00	0.00	60.00	0.00	48.00	108.00

Note. The refundable rental deposit of SGD 99 represents temporary liquidity. Rental travel comprises 24 paid legs at SGD 2 per leg.

rental (Table 3). Second-hand purchasing saves SGD 11.01 relative to new. Rental payments total SGD 60 and travel adds SGD 48. Active time is 120 minutes for each purchase option and 780 minutes for rental, of which 720 minutes arise from dedicated travel.

4.3 Consistency diagnostics and preference comparisons

Under the baseline ratio mapping and $RI = 0.52$, two of 17 individual matrices satisfy $CR \leq 0.10$. Their CR values are zero; nine respondents have CR approximately 0.1304, five have 0.2834, and one has 0.5393. The two qualifying respondents, R006 and R015, each select the midpoint for all three comparisons and therefore each receive equal criterion weights. Their shared result coincides numerically with the normative equal-weight reference.

The full-sample AIJ matrix yields $\lambda_{\max} = 3.013521$, $CI = 0.006760$, and $CR = 0.013001$. This group statistic describes coherence after aggregation. The individual diagnostics characterize the substantial within-respondent inconsistency in the baseline mapping. Together they support an exploratory interpretation of the survey-derived priorities.

The baseline full-sample weights are 0.2009 for emissions, 0.4793 for cost, and 0.3198 for convenience. The milder ratio mapping increases individual consistency acceptance to 11 of 17 and yields weights of 0.2410, 0.4295, and 0.3295. Its group CR is 0.005885. AIP under the baseline mapping yields 0.2027, 0.4663, and 0.3310. Using $RI = 0.58$ retains the same acceptance counts of two and eleven for the respective mappings. These comparisons distinguish sensitivity to ratio intensity from sensitivity to the aggregation stage.

4.4 Composite scores and baseline dominance

Table 5 presents annual outcomes and scores using the baseline full-sample AIJ weights. Second-hand ranks first at 0.8421, followed by new at 0.8274 and rental at 0.5300. The same ordering

Table 4: Criterion weights under alternative mappings and aggregation procedures.

Method	n	Passed	w_E	w_C	w_V	Group CR
Main AIJ	17	2	0.201	0.479	0.320	0.0130
Mild AIJ	17	11	0.241	0.430	0.329	0.0059
Main AIP	17	2	0.203	0.466	0.331	n.a.
Equal weights	—	—	0.333	0.333	0.333	n.a.
Mild AIJ, screened	11	11	0.228	0.422	0.349	0.0038

Note. AIJ aggregates judgments geometrically; AIP averages individual priorities arithmetically. Passed counts individual matrices with $CR \leq 0.10$ under the corresponding mapping. CR values use $RI = 0.52$. The equal-weight reference also matches the two baseline CR-qualified respondents. The entry n.a. denotes a priority-vector specification, for which a group judgment matrix is outside the aggregation procedure.

holds under the milder mapping, AIP, equal weights, and the milder-mapping qualified subset.

Table 5: Baseline annual outcomes, criterion scores, and composite ranking.

Option	kg CO_2e/year	SGD/year	Minutes	s_E	s_C	s_V	S (rank)
Second-hand	4.92	60.99	120	0.7950	0.8306	0.8889	0.8421 (1)
New	4.92	72.00	120	0.7950	0.8000	0.8889	0.8274 (2)
Rental	11.38	108.00	780	0.5258	0.7000	0.2778	0.5300 (3)

Note. Time values are assumed active minutes. Calculations retain full precision; display precision supports comparison of closely spaced scores.

The ordering follows directly from the baseline performance relationships. Second-hand and new have equal modelled emissions and active time, with second-hand costing SGD 11.01 less. New has lower burdens than rental on all three criteria. The corresponding composite-score differences are

$$S_S - S_N = \frac{11.01}{360} w_C, \tag{13}$$

$$S_N - S_R = 0.26919 w_E + 0.10000 w_C + 0.61111 w_V. \tag{14}$$

Both expressions are positive for positive criterion weights. The baseline ordering therefore extends throughout the interior of the normalized weight simplex. At $w_C = 0$, new and second-hand share the same score. This dominance property explains the stability of baseline ranking across preference specifications and identifies operational assumptions as the central drivers of potential ranking changes.

4.5 Transport attribution and ranking crossovers

With the baseline survey weights, the rental score varies as

$$S_R(\alpha) = 0.864748522 - 0.334752218\alpha. \tag{15}$$

At $\alpha = 0$, rental emissions, expenditure, and active time are 4.49 kg CO_2e, SGD 60, and 60 minutes, respectively. The rental score is 0.8647. As the attributed travel share increases, rental first intersects second-hand at $\alpha = 0.067696$ and then new at $\alpha = 0.111488$ (Figure 3). Thus rental ranks first below approximately 6.77%, second between approximately 6.77% and 11.15%, and third above approximately 11.15%. Each crossover assigns equal scores to the intersecting options.

The rental–second-hand crossover is approximately 6.87% under the milder full-sample mapping, 6.83% under AIP, and 7.00% under equal weights (Table 6). These values indicate a closely grouped first-place crossover across the evaluated preference specifications. The transport attribution assumption determines the operational scenario, while preference weights adjust the precise crossover location.

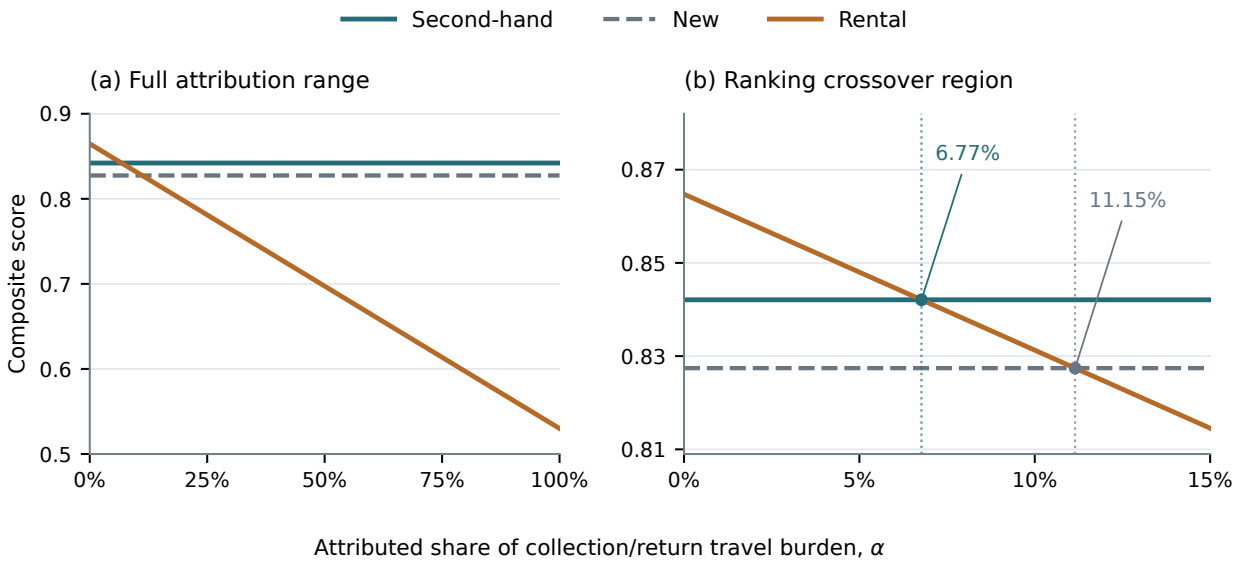


Figure 3: Composite-score sensitivity to the share of rental acquisition-travel burdens attributed to rental. The full range connects the shared-journey boundary to the dedicated-travel baseline; the enlarged crossover region resolves changes in ordering. Both panels use the original 17-response baseline AIJ weights.

Table 6: Rental crossover shares under alternative criterion-weight specifications.

Weight specification	Rental = second-hand	Rental = new
Main AIJ	6.77%	11.15%
Mild AIJ	6.87%	10.67%
Main AIP	6.83%	11.01%
Equal weights	7.00%	9.82%
Mild AIJ, screened	6.95%	10.58%

Note. Each entry is the percentage of baseline rental travel emissions, fares, and time at which the indicated pair has equal composite scores. Ownership outcomes and other rental components remain fixed.

5 Discussion

5.1 Mechanisms underlying the acquisition comparison

The baseline advantage of second-hand purchasing arises from its lower acquisition payment under common garment-stage allocation, cleaning, and delivery assumptions. Equal attributed emissions for the two purchase options follow from this modelling convention. The common 26-wear lifetime similarly equalizes rental's allocated garment burden with ownership. The analysis therefore isolates service operations and annual payments at a common lifetime utilization level.

Rental's low booking price combines with repeated travel payments and time commitments over six events. The transport analysis quantifies this mechanism: a rental advantage emerges near the lower boundary of incremental acquisition travel. The small first-place crossover share, approximately 6.8% under the baseline weights, indicates the extent of travel sharing associated with that advantage within the proportional scenario model. Practical relevance depends on actual trip chaining, routes, fares, and timing. These relationships motivate service arrangements that integrate collection and return with existing journeys.

5.2 Interpretation of the preference evidence

The survey provides two complementary forms of evidence. Question-level counts describe expressed preference directions, including the predominance of cost over emissions in Q8. AHP translates those responses into exploratory numerical priorities under explicit ratio and aggregation assumptions. The personal CR distribution and variation across mappings indicate sensitivity in the numerical strength assigned to the recorded categories.

The increase in acceptance under the milder mapping reflects a change in the analytical conversion. The AIP comparison assesses how the aggregation stage influences collective priorities. The two-person qualified subset illustrates the selection effect of the baseline consistency threshold, with its all-midpoint responses producing the equal-weight vector. These results preserve the distinction between preference description, matrix consistency, and population inference. Baseline ranking stability follows from criterion dominance, while the transport crossovers provide a separate assessment of ranking sensitivity under changing service conditions.

5.3 Evidence scope and further analytical development

The environmental inventory combines garment proxies, rounded graphical extractions, an Italian residential-care estimate, and UK transport factors. Detailed and overview source plots exhibit reconciliation differences, and the selected factors retain their original methodological conventions. Product-specific inventories and harmonized local factors would strengthen the precision and geographical applicability of the estimated burdens. Different channel-specific lifetimes and alternative second-hand allocation conventions represent additional dimensions for scenario analysis.

The financial comparison uses advertised quotes and annual cash expenditure with zero resale and residual values. Its interpretation concerns cash payments during the study year. Transaction prices, service availability, care arrangements, and year-end garment value would support broader economic characterization. The refundable deposit describes a distinct liquidity requirement.

Convenience is represented by assumed active minutes. Search, fitting, availability management, and broader effort contribute further dimensions to this construct. The fixed scoring anchors also shape trade-offs through their interaction with criterion weights. The transport scenario offers an analytically transparent proportional relationship among incremental emissions, fares, and time; route-specific evidence would permit separate variation of these components. The 17 complete comparison sets support exploratory preference analysis, with broader inference requiring additional evidence on respondent coverage and elicitation quality.

6 Conclusion

For the six-event baseline, second-hand purchasing produces the highest composite score, followed by new purchasing and rental. Both purchase options yield 4.92 kg CO_2e/year, compared with 11.38 kg CO_2e/year for rental. Annual expenditure is SGD 60.99 for second-hand, SGD 72.00 for new, and SGD 108.00 for rental. The second-hand advantage over new is attributable to its SGD 11.01 cost saving under equal modelled emissions and active time.

Original questionnaire responses support an exploratory assessment of criterion preferences. Two of 17 individual matrices meet the baseline consistency criterion, and alternative mappings and aggregation procedures provide transparent sensitivity comparisons. The baseline ranking follows from the alternatives' criterion-level dominance across positive weight settings. Under the selected score anchors and proportional transport scenario, rental becomes the highest-scoring option below approximately 6.77% of baseline rental travel burdens, with the evaluated preference specifications placing the first-place crossover near 7%. The study consequently links the acquisition recommendation to garment allocation, service operations, annual payments, and the treatment of collection-and-return travel.

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Bibliographic note. Citation years follow the proceedings volume for Saaty (1994) and the journal volume for Monticelli and Costamagna (2023). Their online publication dates are June 1, 2009 and 2022, respectively. The original survey and analytical outputs are documented in Appendix A.

A Source and Reproduction Notes

A.1 Market and environmental evidence

Market sources are indexed as M2_001 (new), M2_002 (second-hand listing 1447640322), and M2_003 (rental listing 1464833759) in the source register, with access dated 3 November 2026, Singapore time. The baseline uses the first-rental refundable deposit of SGD 99. Literature locators are Sandin et al. (2019), printed page 83, Figure 4.24, for garment-stage burdens and printed page 159, Table B-59, for the 0.478 kg reference mass. The cleaning proxy is the residential GWP entry in Monticelli and Costamagna (2023), Table 4, printed page 7635; Table 2 on page 7632 describes the associated inventory.

UK 2026 rail factors comprise direct and upstream terms of $0.02121 + 0.00749 = 0.02870$ kg CO_2e/passenger-km from *Business travel–land* D89 and *WTT–pass vehs & travel–land* D86. Van factors comprise $0.25716 + 0.06128 = 0.31844$ kg CO_2e/vehicle-km from *Delivery vehicles* D32 and *WTT–delivery vehs & freight* D31. The source evidence archive contains selected cells and file hashes.

A.2 Survey and analytical traceability

The original workbook's response sheet, rows 2–19 and columns J–L, supplies Q8–Q10. The question-and-rules sheet provides the archived wording. Q9 endpoints run from convenience to emissions; Q10 endpoints run from cost to convenience. Intermediate positions are numbered, and the analysis assigns symmetric ratios around code 3. Q8 defines cost net of resale or retained value; the baseline zero-value assumptions align that quantity with annual cash expenditure.

The revised analysis archive records question-level counts, individual consistency diagnostics, five weight specifications, annual burdens, transport equations, and crossover shares in `analysis/results.json`. The script `analysis/recompute.py` recreates the analysis from the original workbook and project parameters. Source hashes identify the inputs. The standalone LaTeX package includes the generated tables and vector figures required for compilation. Numerical comparisons use full-precision values, with rounded presentation in the main text.

A.3 Authorship

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AI use statement. AI assistance supported English drafting and revision, LaTeX preparation, figure generation, and computational and document checks.